

Full Stack Development with MERN Project Documentation

1. Introduction :

Project Title: Book Store – MERN Stack Web Application

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2. Project Overview

Purpose:

The Book Store is a full-stack web application developed using the MERN stack. It provides an online platform where users can browse books, register, log in, search books, manage their profiles, and purchase books. The application also provides separate dashboards for Admin and Sellers to efficiently manage books, users, and orders.

Key Features:

User Module

- User Registration
- User Login
- JWT Authentication
- Search Books
- Browse Categories
- Book Details
- Shopping Cart
- Place Orders
- Profile Management

Seller Module

- Seller Registration/Login
- Add Books
- Update Books
- Delete Books
- Manage Inventory
- View Orders

Admin Module

- Admin Login
- Dashboard
- Manage Users
- Manage Sellers
- Manage Books
- Monitor Orders
- View Platform Statistics

3. Architecture

Frontend

Technology Used

- React.js
- React Router
- Axios
- CSS
- Bootstrap

Responsibilities

- Responsive UI
- Authentication Pages
- Dashboard
- Book Listing

- Shopping Cart
 - Profile Pages
 - Admin Panel
 - Seller Panel
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Backend

Technology Used

- Node.js
- Express.js

Responsibilities

- REST APIs
 - Authentication
 - Authorization
 - CRUD Operations
 - Order Processing
 - Book Management
 - User Management
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Database

Technology Used

- MongoDB
- Mongoose

Collections

- Users
- Admin
- Sellers
- Books
- Orders

4. Setup Instructions

Prerequisites

Install the following software:

- Node.js (v16 or above)
- npm
- MongoDB
- Git
- VS Code

Installation

Move into project

```
cd Book-Store
```

Install Backend Packages

```
cd server
```

```
npm install
```

Install Frontend Packages

```
cd client
```

```
npm install
```

Configure Environment Variables

Create a .env

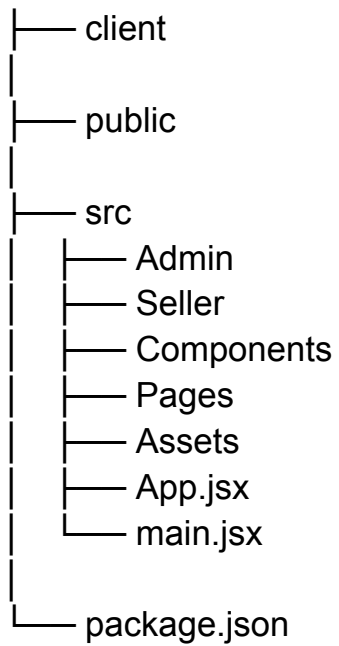
```
PORT=5000
```

```
MONGO_URI=your_mongodb_connection_string
```

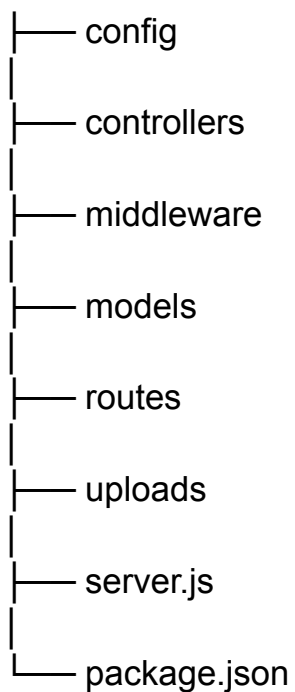
JWT_SECRET=your_secret_key

5. Folder Structure :

Book-Store



server



6. Running the Application

Backend

cd server

npm start

or

npm run dev

Frontend

cd client

npm start

or

npm run dev

Open

<http://localhost:5173>

7. API Documentation

Authentication APIs

Method	Endpoint	Description
POST	/register	Register User
POST	/login	User Login

Book APIs

Method	Endpoint	Description
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GET	/books	Get Books
GET	/books/:id	Get Book Details
POST	/books	Add Book
PUT	/books/:id	Update Book
DELETE	/books/:id	Delete Book

User APIs

Method	Endpoint
GET	/users
PUT	/users/:id
DELETE	/users/:id

Seller APIs

Method	Endpoint
POST	/seller/login
POST	/seller/register
GET	/seller/books

Order APIs

Method	Endpoint
POST	/orders
GET	/orders

8. Authentication

Authentication is implemented using **JWT (JSON Web Token)**.

Workflow:

- User logs in.
- Server validates credentials.
- JWT token is generated.
- Token is stored on the client.
- Protected APIs verify the token.
- Unauthorized users cannot access restricted routes.

Authorization Levels

- Admin
- Seller
- user

9. User Interface

The application provides a modern and responsive interface.

Main Pages

- Home Page
- Sign up
- Book Listing
- Book Details
- Cart
- Orders
- User Dashboard
- Seller Dashboard
- Admin Dashboard

10. Testing

Testing Performed

- User Registration Testing
- Login Authentication

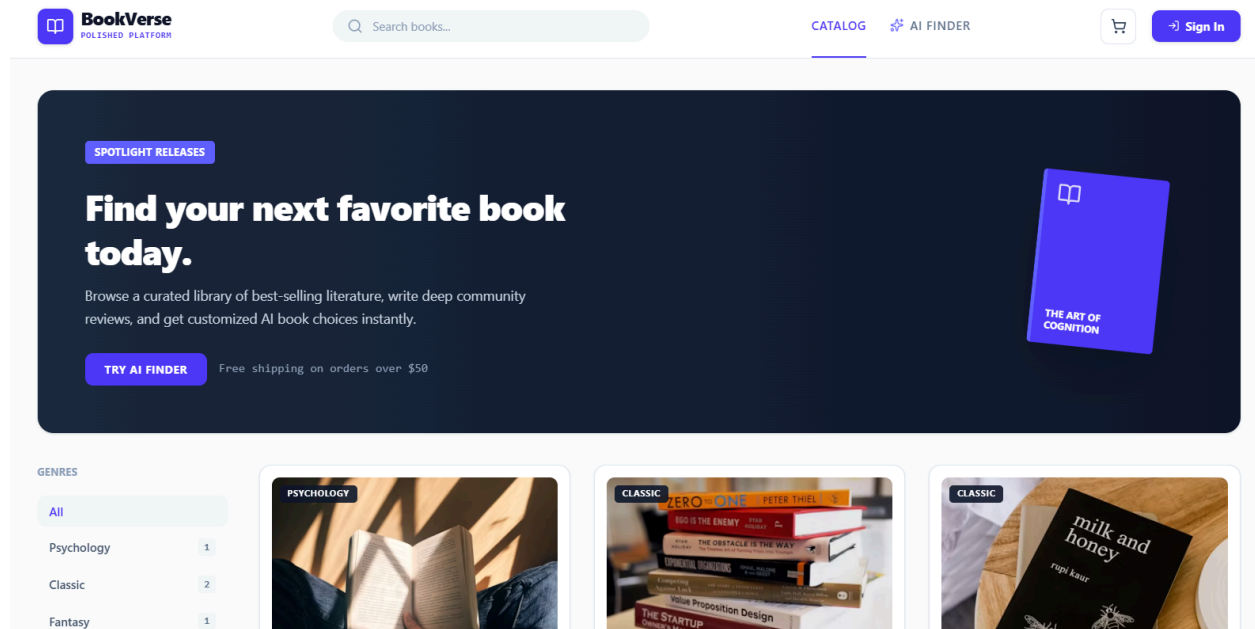
- CRUD Operations
- API Testing
- Responsive UI Testing
- MongoDB Connection Testing
- Form Validation
- Error Handling

Tools Used

- Postman
- Browser Developer Tools
- MongoDB Compass

11. Screenshots / Demo

Home Page



Librarian AI Finder

Describe your preferred reading style or select a genre, and our Gemini-powered AI engine will generate personalized book recommendations suited exactly to your vibe.

SELECT GENRE CATEGORY

All Genres

Sci-Fi

Classic

Fantasy

Self-Help

Psychology

Mystery

Biography

Business

DESCRIBE YOUR READING VIBE / PREFERENCE

E.g., An immersive fantasy world with deep political lore and magic system, or an encouraging guide to business startups...

OR TRY ONE OF THESE SUGGESTIONS:

A fast-paced mystery with a major plot twist

Deep philosophical thoughts and slow-burn character growth

Inspiring guide to building positive mental routines

Immersive futuristic science fiction with advanced technologies

Warm and comforting historical romance

🧠 GENERATE BOOK RECOMMENDATIONS



YOUR SHOPPING CART



Atomic Habits
by James Clear

\$16.99

\$16.99 each

- 1 +



Subtotal **\$16.99**

Estimated Tax (8%) **\$1.36**

Shipping **\$5.99**

Add \$33.01 more for free shipping!

Grand Total \$24.34



DELIVERY & SHIPPING

12. Known Issues

- Payment gateway integration is not yet implemented.
- Email notifications are currently unavailable.
- Advanced analytics dashboard can be improved.
- Wishlist functionality is under development.

13. Future Enhancements

- Online Payment Integration
- AI Book Recommendation System
- Book Rating & Reviews
- Wishlist
- Order Tracking
- Email Notifications
- Chat Support
- Multi-language Support
- Dark Mode
- Mobile Application
- Cloud Deployment
- Inventory Analytics Dashboard